

Software Development Group Project Proposal Presentation **SMART PMB**



Group No: D
Digital Paddy Ecosystem for
Sri Lanka

TEAM MEMBERS



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INTRODUCTION



AI-powered digital ecosystem to modernize Paddy Marketing Board (PMB) operations in Sri Lanka.

Background:

- Smart PMB is an AI-powered digital ecosystem for Sri Lanka's Paddy Marketing Board.
- PMB currently relies on manual paperwork, disconnected workflows, and limited digital tools.
 - Farmers struggle with delayed payments, lack of price information, and poor communication.
 - PMB officers face challenges in stock monitoring, warehouse management, and analytics.
- Smart PMB integrates mobile apps, web dashboards, AI analytics, and cloud infrastructure.
- Aims to improve transparency, operational efficiency, and data-driven decision-making.



OBJECTIVES

Main Objective: Develop an intelligent centralized platform to digitize PMB operations with AI-powered analytics and role-based management.

- To digitize paddy purchasing operations and automate transaction workflows.
- To improve farmer accessibility through mobile technology and real-time notifications.
- To implement centralized warehouse and stock management across all districts.
- To integrate AI-powered price prediction, yield forecasting, and crop disease detection.



SCOPE OF THE PROJECT



DELIVERABLES

Deliverables of the Smart PMB system:

- Smart PMB Web-Based Platform
- AI-Powered Analytics Modules
- Centralized Database & Secure Backend
- QR-Based Digital Transaction System
- Integrated Warehouse & Logistics System
- Full Testing & Validation Reports
- Deployment & Technical Documentation



SCOPE OF THE PROJECT



SYSTEM CONSTRAINTS

The following are out of scope for the Smart PMB system:

- Limited to domestic PMB operations within Sri Lanka
- No integration with international export systems
- No direct banking or taxation system integration
- No satellite, drone, or IoT-based monitoring
- Web-based platform only (no hardware deployment)
- Future technologies such as blockchain are excluded

FUNCTIONAL REQUIREMENTS

- *System-wide functional capabilities of Smart PMB:*
 - *Role-based functions assigned to each user type:*
-
- Farmer: Register profile, upload harvest details, view guaranteed prices
 - Track paddy sales and payment history
 - Receive real-time notifications from PMB officers
 - Authorized Purchaser: Record purchases, verify transactions, manage stock intake
 - PMB Officer: Monitor national operations, approve licenses, manage warehouses

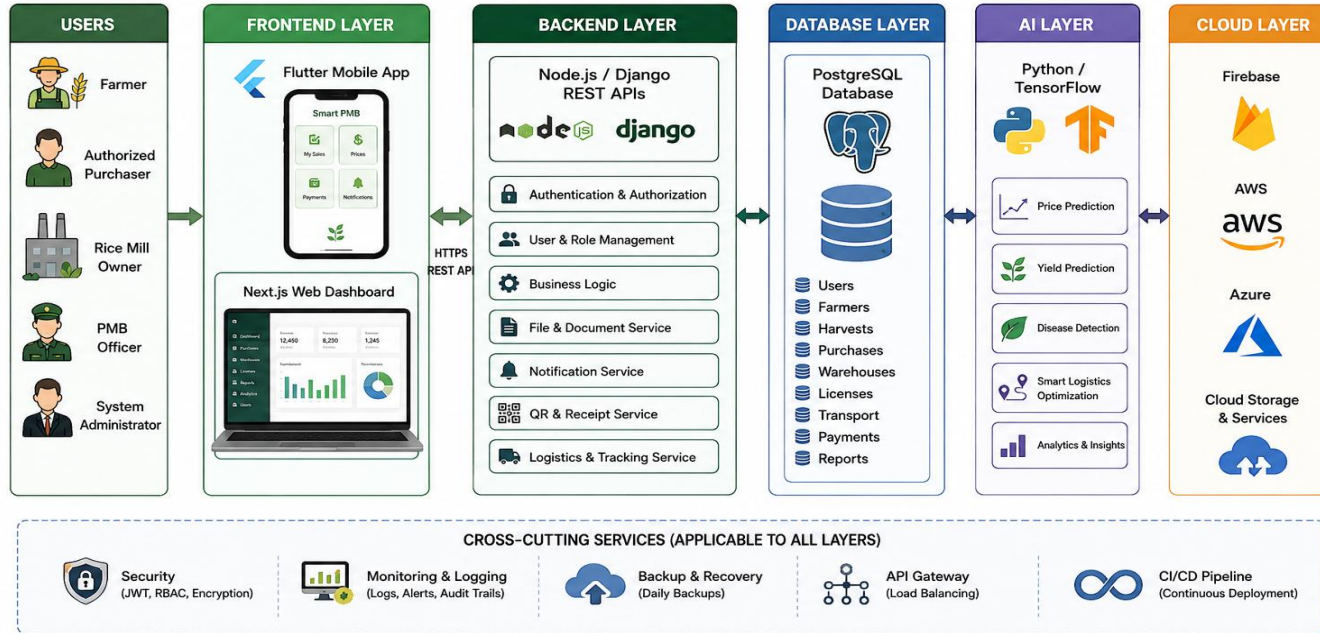
NON FUNCTIONAL REQUIREMENTS

Non-functional requirements for Smart PMB:

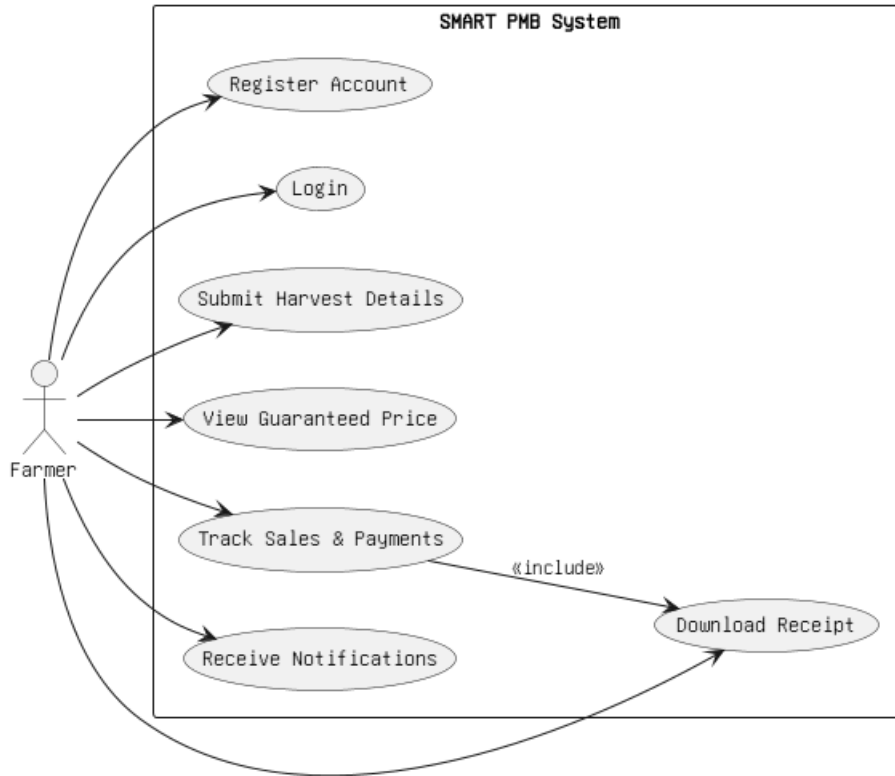
- *Performance: Fast response times with real-time synchronization and efficient database queries.*
- *Security: JWT authentication, OTP verification, encrypted passwords, secure APIs, and audit logging.*
- *Usability: Mobile-friendly interface with Sinhala, Tamil, and English language support.*
- *Reliability: Daily backups and disaster recovery systems to ensure continuous availability.*
- *Scalability: Multi-district support with cloud scalability to handle nationwide operations.*
- *Maintainability: Modular architecture enabling easy updates and feature enhancements.*
- *Compatibility: Responsive design working across web browsers and Android/iOS mobile devices.*

SYSTEM ARCHITECTURE

Smart PMB uses a layered architecture: Flutter Mobile + Next.js Web (Frontend) → Node.js/Django REST APIs (Backend) → PostgreSQL (Database) → Python/TensorFlow AI Layer → Firebase/AWS/Azure (Cloud)



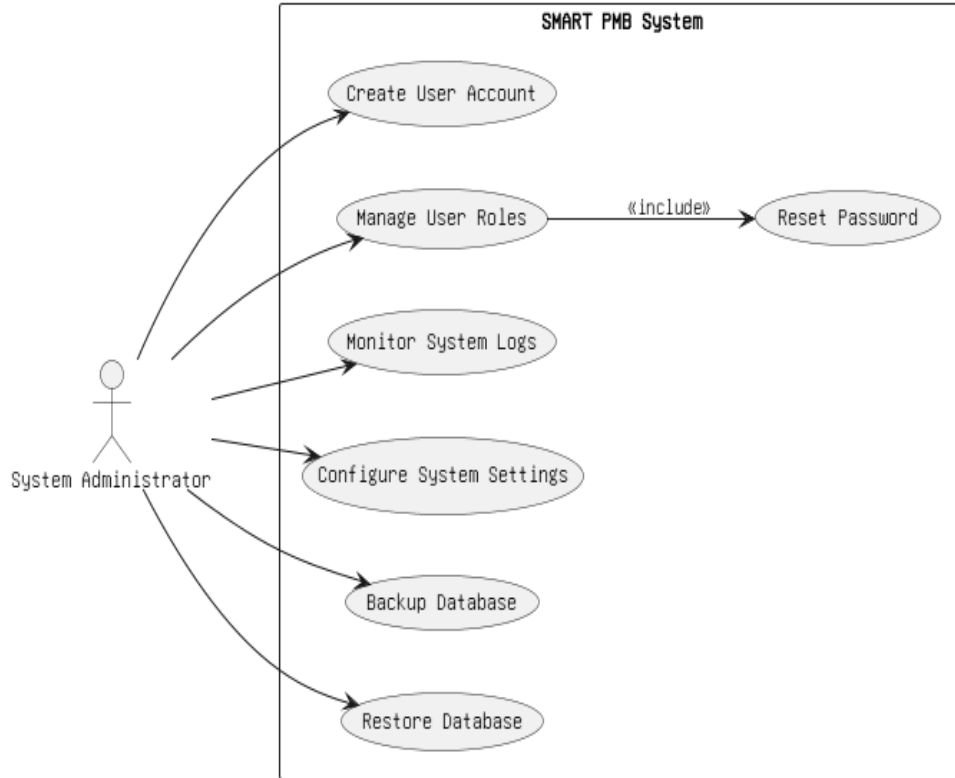
USE CASE DIAGRAM



Farmer

- The **Farmer** interacts directly with the Smart PMB System.
- The **"Track Sales & Payments"** use case includes **"Download Receipt"**, meaning receipt downloading is part of viewing transaction history.
- All activities are performed through the web-based dashboard.

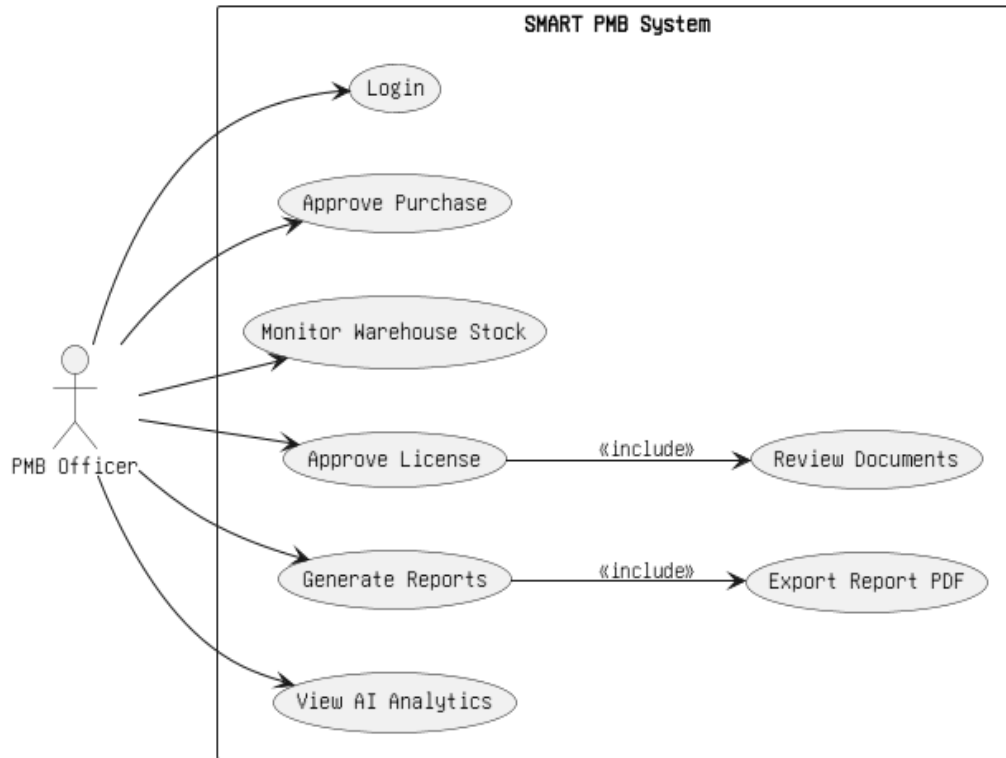
USE CASE DIAGRAM



System Administrator

- **Manage User Roles** includes **Reset Password**.
- Admin ensures secure and stable operation of the platform.

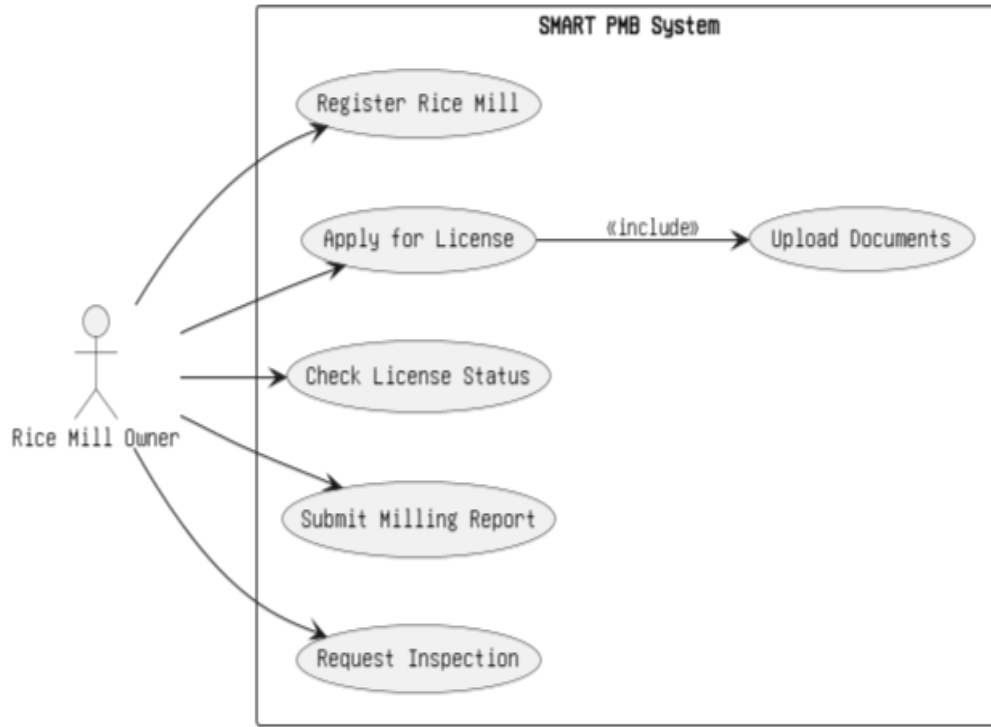
USE CASE DIAGRAM



PMB Officer

- **Approve License** includes **Review Documents**.
- Officers use AI-generated insights for decision-making.

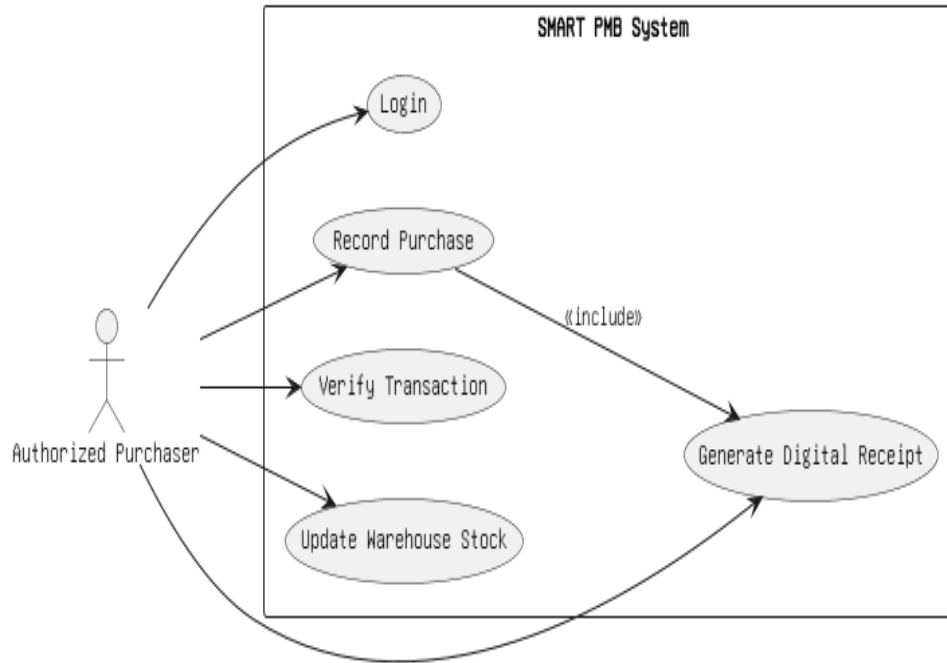
USE CASE DIAGRAM



Rice Mill owner

- The **Apply for License** use case includes **Upload Documents**.
- The PMB Officer reviews and approves license applications.

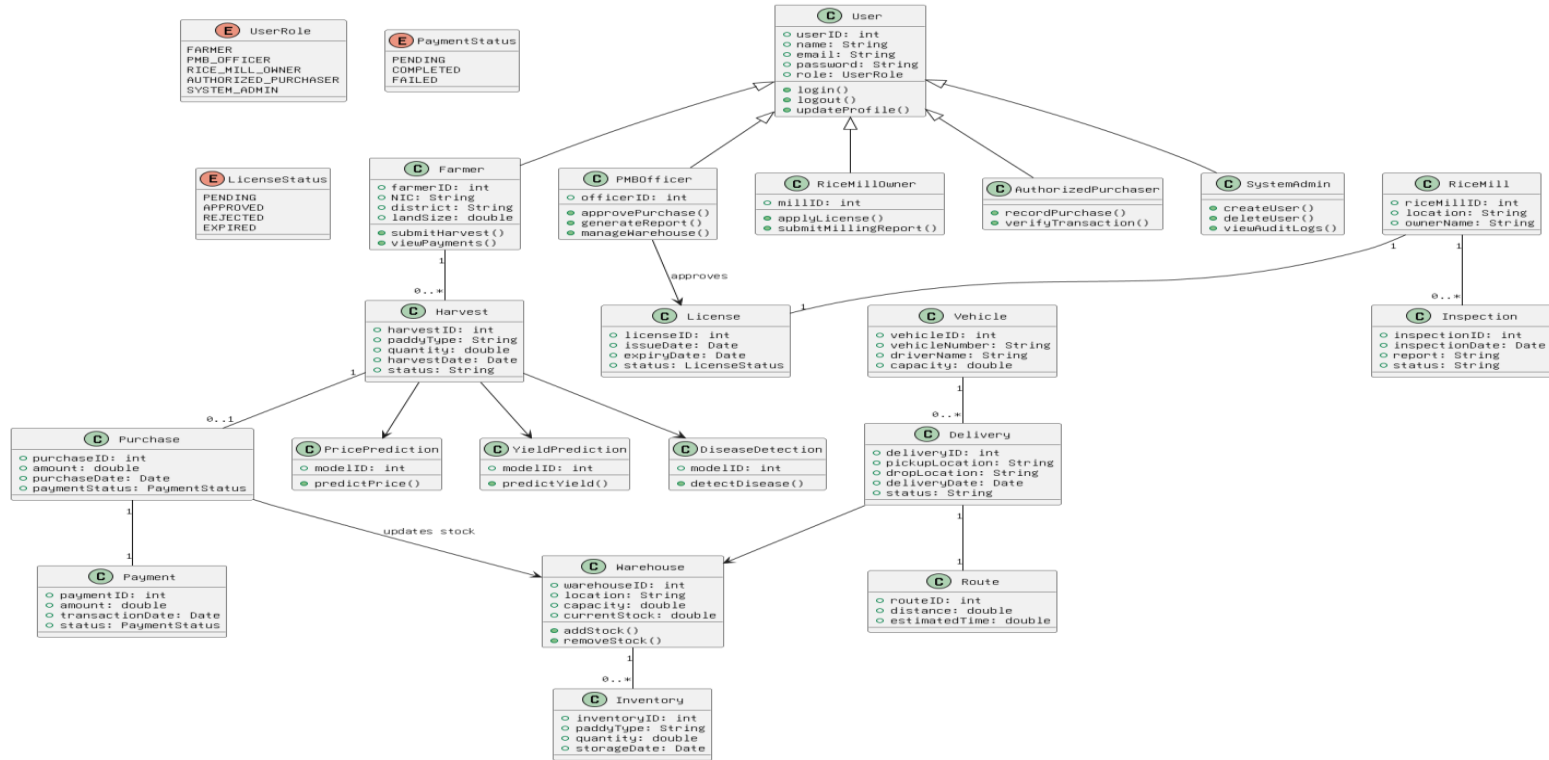
USE CASE DIAGRAM



Authorized purchaser

- The **Record Purchase** use case includes **Generate Digital Receipt**.
- All purchase activities update the centralized database.
- The system ensures secure transaction recording and stock management.

CLASS DIAGRAM



SOFTWARE EVALUATION AND TESTING

The following testing strategy will be applied to Smart PMB:

- Unit Testing – Testing individual modules (Farmer, Warehouse, AI) independently.
- Integration Testing – Verifying communication between frontend, backend, and AI layers.
- System Testing, User Acceptance Testing (UAT), and Security Testing for authentication and data protection.

PROJECT LIMITATIONS AND CHALLENGES

Key limitations and challenges anticipated in the Smart PMB project:

- Time limitation – 7 sprint cycles within the academic timeline.
- Learning curve – AI/ML integration (TensorFlow, Scikit-learn) and Flutter mobile development.
- Data availability – Obtaining real paddy price and yield datasets for AI model training.



TECHNOLOGY STACK

DESIGNING TOOLS (UI/UX)



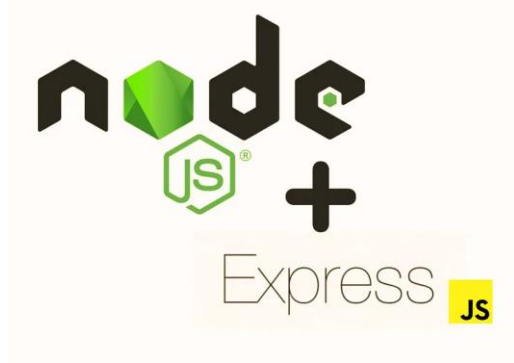
TECHNOLOGY STACK

FRONT-END TECHNOLOGIES



TECHNOLOGY STACK

BACK-END (SERVER-SIDE)



TECHNOLOGY STACK

DATABASE

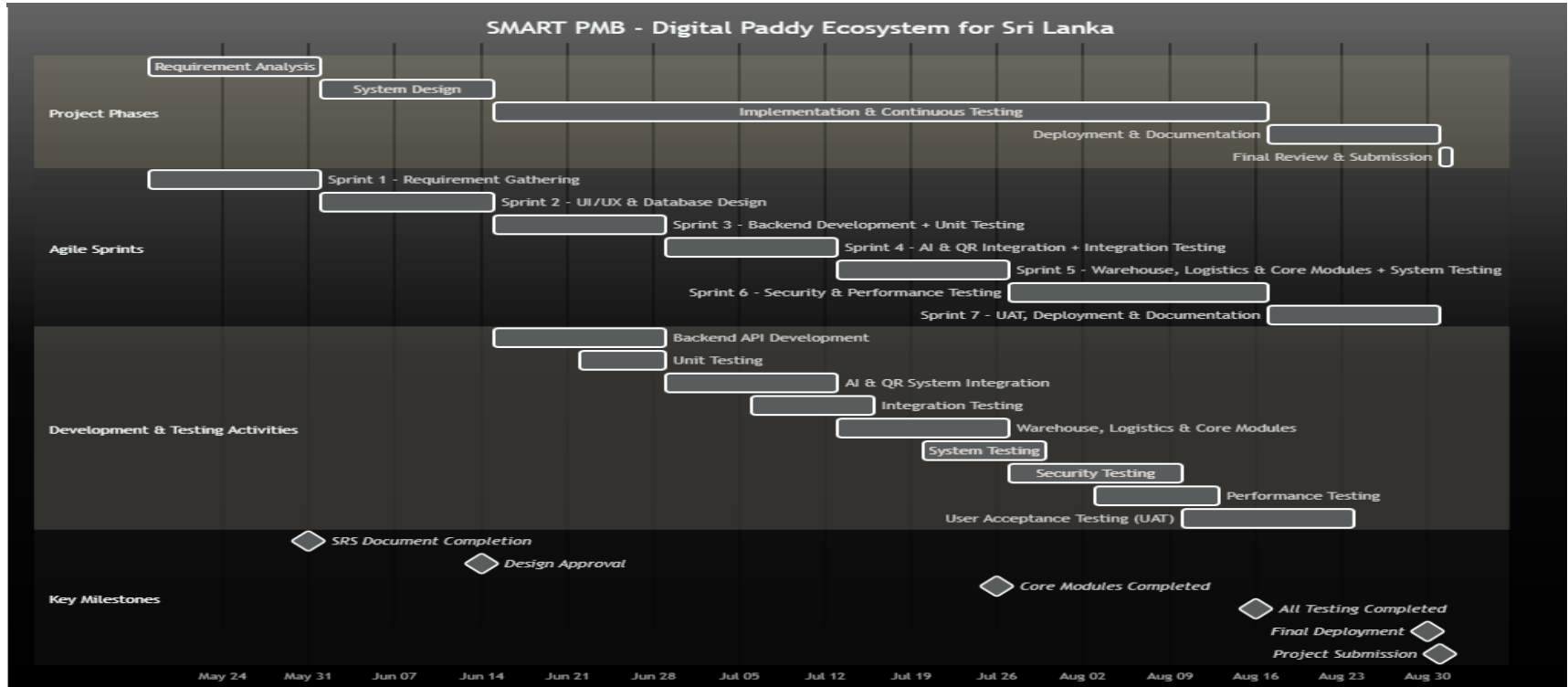


VERSION CONTROL

GitHub (version control, branching strategy, CI/CD pipeline)



GANTT CHART



THANKYOU !!

Q & A